

Nurture Course

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2023

Test Type : **PRACTICE TEST**

Test Pattern : **NEET (UG)**

TEST DATE :

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	2	2	2	3	1	3	3	4	2	4	3	1	3	2	1	2	4	3	1	1	2	2	1	1	2	4	2	2	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	1	2	3	2	1	3	2	1	4	3	3	2	3	1	2															

HINT - SHEET



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1. **Ans (3)**

$\therefore mgh = \text{constant}$

$$g_A h_A = g_B h_B \Rightarrow h_B = \frac{g_A}{g_B} \times h_A = \frac{6}{1} \times 2 = 12 \text{ m}$$

2. **Ans (2)**

Gravitational force does not depend on medium.

3. **Ans (2)**

$$g = \frac{GM}{R^2} = \frac{G \frac{4}{3} \pi R^3 \rho}{R^2} = \frac{4}{3} \pi G \rho R$$

$$\Rightarrow r \propto R \quad g_p = g_e$$

$$\rho_p R_p = \rho_e R_e$$

$$R_p = \frac{\rho_e R_e}{\rho_p} = \frac{\rho R}{3\rho} = \frac{R}{3}$$

4. **Ans (2)**

$$U = -\frac{GMm}{r}$$

$$K = \frac{GMm}{2r}$$

$$\frac{U}{K} = 2$$

5. **Ans (3)**

$$U = -\frac{GMm}{r}, K = \frac{GMm}{2r}$$

$$E = K + U = -\frac{GMm}{r} + \frac{GMm}{2r} = -\frac{GMm}{2r}$$

$$U = 2E_0$$

6. **Ans (1)**

by conservation of mechanical energy

$$-\frac{GMm}{R+2R} + \frac{1}{2}m(fv)^2 = 0$$

$$\Rightarrow (fv)^2 = \frac{2GM}{3R} \Rightarrow fv = \sqrt{\frac{2GM}{R}} \times \frac{1}{\sqrt{3}}$$

$$\Rightarrow fv = \frac{1}{\sqrt{3}} v_{es} \rightarrow f = \frac{1}{\sqrt{3}}$$

7. **Ans (3)**

Time period of a satellite is independent of its mass.

8. **Ans (3)**

$$V_0 = \sqrt{\frac{GM}{r}}$$

$$\frac{V_{02}}{V_{01}} = \sqrt{\frac{r_1}{r_2}} = \sqrt{\frac{R}{R+3R}} = \frac{1}{2}$$

$$\Rightarrow V_{02} = 0.5 \times V_0$$

9. **Ans (4)**

Areal velocity is constant so

$$\Rightarrow \frac{A_{SCD}}{t_1} = \frac{A_{SAB}}{t_2} \Rightarrow \frac{3A_{SAB}}{t_1} = \frac{A_{SAB}}{t_2}$$

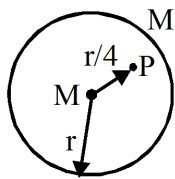
$$\therefore t_1 = 3t_2$$

10. **Ans (2)**

$$\text{Speed of satellite } V = \sqrt{\frac{GM}{r}}$$

$$\Rightarrow \frac{V_B}{V_A} = \sqrt{\frac{r_A}{r_B}} = \sqrt{\frac{9R}{R}} = 3 \Rightarrow V_B = 3V_A = 3V$$

11. Ans (4)



$$V_P = -\frac{GM}{r/4} - \frac{GM}{r} = -\frac{5GM}{r}$$

12. Ans (3)

Gravitational field inside hollow shell is zero. Hence gravitational field at A will be due to inner shell only.

13. Ans (1)

Inside $E = 0$

outside $E \propto \frac{1}{r^2}$

14. Ans (3)

$$E_i = -\frac{GM_em}{4R_e}$$

$$E_f = -\frac{GM_em}{8R_e}$$

$$\Delta E = E_f - E_i = \frac{GM_em}{8R_e}$$

15. Ans (2)

$$U_i = -\frac{GMm}{3R}$$

$$U_f = -\frac{GMm}{2R}$$

Loss in PE = gain in KE

$$\Delta U = U_f - U_i = -\frac{GMm}{6R}$$

16. Ans (1)

During path DAB planet is nearer to sun as compared to path BCD. Velocity during path DAB will be more hence time taken will be less.

17. Ans (2)

By conservation of angular momentum

$$v_1 r_1 = v_2 r_2$$

$$v_1 a(1-e) = v_2 a(1+e)$$

$$\frac{v_1}{v_2} = \frac{1+e}{1-e} = 2 \Rightarrow e = \frac{1}{3}$$

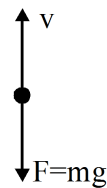
18. Ans (4)

$$P = \vec{F} \cdot \vec{v}$$

At the time of projection

Angle between $\vec{F}$ and $\vec{v}$ is 180°

hence power is lowest



19. Ans (3)

Potential energy at height h

$$U = -\frac{mgR}{1 + \frac{h}{R}} \quad \text{at } h = 4R$$

$$U = -\frac{mgR}{1 + 4} = -\frac{mgR}{5}$$

So minimum energy required to escape the body will be $\frac{mgR}{5}$.

20. Ans (1)

$$g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$$

$$\frac{g}{4} = \frac{g}{\left(1 + \frac{h}{R}\right)^2} \Rightarrow \frac{1}{2} = \frac{1}{1 + \frac{h}{R}} \Rightarrow h = R$$

21. Ans (1)

$$T = \frac{2\pi R^{3/2}}{\sqrt{GM}} \Rightarrow T^2 = \frac{4\pi^2 R^3}{GM}$$

22. Ans (2)

$$\text{on earth } g_e = \frac{GM}{R^2}$$

$$\text{on moon } g_m = \frac{GM}{80\left(\frac{R}{4}\right)^2} = \frac{GM}{5R^2}$$

$$g_m = \frac{g}{5}$$

23. Ans (2)

$$v_e = \sqrt{2gR} \text{ and } v_0 = \sqrt{gR}$$

$$\% \text{ change} = \frac{v_e - v_0}{v_0} \times 100\%$$

$$\frac{(\sqrt{2} - 1)}{\sqrt{gR}} \times \sqrt{gR} \times 100\% = 41\%$$

24. Ans (1)

$$\frac{1}{2} m v_\infty^2 = \frac{1}{2} m (2V_e)^2 - \frac{1}{2} m V_e^2$$

$$\Rightarrow v_\infty^2 = 3V_e^2 \Rightarrow v_\infty = \sqrt{3}V_e$$

25. Ans (1)

$$\Delta U = \frac{mgh}{1 + \frac{h}{R}}$$

$$\Delta U = \frac{mg \frac{R}{2}}{1 + \frac{R}{2R}} = \frac{1}{3}mgR$$

26. Ans (2)

$$V = -G(1) \left[\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right]$$

$$V = -G(1) \frac{1}{1 - \frac{1}{2}}$$

$$V = -2G$$

27. Ans (4)

$$V_e = \sqrt{\frac{2GM}{R}} \Rightarrow M = \frac{V_e^2 R}{2G}$$

$$= \frac{(3 \times 10^8)^2 \times 10^{-2}}{2 \times 6.67 \times 10^{-11}} = 6.7 \times 10^{24} \text{ kg}$$

$$\simeq 10^{24} \text{ kg}$$

28. Ans (2)

$$V_0 = \sqrt{\frac{GM}{R+h}} = \sqrt{\frac{GM}{R \left(1 + \frac{h}{R}\right)}} = \sqrt{\frac{gR}{1 + \frac{h}{R}}}$$

$$V_0 = \sqrt{\frac{9.8 \times 6.4 \times 10^6}{1 + \frac{1.6 \times 10^6}{6.4 \times 10^6}}} = 7.1 \text{ km/s}$$

29. Ans (2)

$$v = \sqrt{\frac{GM}{r}} \text{ orbital speed does not depend on}$$

mass of the satellite $v \propto \frac{1}{r}$ so if $r_1 > r_2$, then $v_1 < v_2$

30. Ans (2)

$$v_0 = \sqrt{\frac{GM}{r}} \quad v_0 \propto \frac{1}{r}$$

we need more than one satellite for global communication for stable orbit, satellite orbit must pass through centre of earth. Only (2) is correct.

31. Ans (1)

Time period and sense of rotation of geostationary satellite must be same as earth.

32. Ans (2)

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2G}{R} \times \frac{4}{3}\pi R^3 \rho}$$

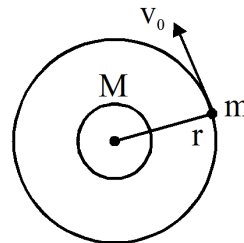
$$v_e \propto R\sqrt{\rho}$$

$$\therefore \frac{v_e}{v_p} = \frac{1}{3\sqrt{3}}$$

33. Ans (3)

$$F_c = \frac{mv_0^2}{r}$$

$$F_g = \frac{GMm}{r^2}$$



34. Ans (2)

Angular momentum of system will be conserved about sun only. Linear momentum will vary as velocity varies only total energy is conserved.

35. Ans (1)

COAM $mvr = \text{constant}$

$$r_C > r_B > r_A$$

$$\therefore v_A > v_B > v_C$$

$$\therefore K_A > K_B > K_C$$

$$\Rightarrow U_A < U_B < U_C$$

36. Ans (3)

$$m'_s = \frac{m_s}{4} \text{ and } G' = 10G$$

$$g_e = \frac{G' M_E}{R^2} = \frac{10GM_E}{R^2} = 10g$$

37. Ans (2)

$v_e < v_{rms}$ O_2 molecule will leave the surface.

38. Ans (1)

$$V = -\frac{GM}{R+h} = -4.0 \times 10^7 \quad \dots(i)$$

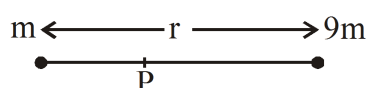
$$g = \frac{GM}{(R+h)^2} = 5 \quad \dots(ii)$$

Solving (i) and (ii) $R+h = 8000$

$$h = 8000 - 6400$$

$$= 1600 \text{ km}$$

39. Ans (4)



$$\frac{Gm}{x^2} = \frac{G9m}{(r-x)^2} \Rightarrow (r-x)^2 = 9x^2$$

$$\Rightarrow r-x = \pm 3x$$

$$\therefore r = 4x \text{ or } r = -2x \text{ (not possible)}$$

$$\therefore x = \frac{r}{4}$$

$$V_P = -\frac{Gm}{\frac{r}{4}} - \frac{G \times 9m}{\frac{3r}{4}} = -\frac{GM}{r}(4+12) = -\frac{16Gm}{r}$$

40. Ans (3)

$$T^2 = \frac{4\pi^2}{GM} r^3$$

$$\Rightarrow r^3 = \frac{GMT^2}{4\pi^2}$$

$$\Rightarrow R+h = \left(\frac{GMT^2}{4\pi^2} \right)^{\frac{1}{3}}$$

$$\Rightarrow R+h = \left(\frac{GMT^2}{4\pi^2} \right)^{\frac{1}{3}}$$

$$\Rightarrow h = \left(\frac{GMT^2}{4\pi^2} \right)^{\frac{1}{3}} - R$$

$$\Rightarrow h = \left(\frac{GMT^2}{4\pi^2} \right)^{\frac{1}{3}} - R$$

41. Ans (3)

Solid sphere $\Rightarrow E_{in} \propto r$

$$\text{and } E_{out} = \frac{GM}{r^2} E_{out} = \frac{GM}{r^2} \Rightarrow E_{out} \propto \frac{1}{r^2}$$

42. Ans (2)

$$F = \frac{GM \cdot M}{d^2} = \frac{GM^2}{d^2}, F' = \frac{G \frac{M}{2} \cdot \frac{3M}{2}}{\left(\frac{d}{2}\right)^2}$$

$$F = \frac{GM \cdot M}{d^2} = \frac{GM^2}{d^2}, F' = \frac{G \frac{M}{2} \cdot \frac{3M}{2}}{\left(\frac{d}{2}\right)^2}$$

$$F' = \frac{3GM}{d^2} = 3F, F' = \frac{3GM}{d^2} = 3F$$

Gravitational force does not depend on medium

43. Ans (3)

$$KE = E = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2E}{m}}$$

$$v = \sqrt{\frac{2E}{m}}$$

$$L = mvr = m \sqrt{\frac{2E}{m}} \cdot r = \sqrt{2Emr^2}$$

44. Ans (1)

Total energy of satellite

$$E = -\frac{GMm}{2(R+h)} = -\frac{GMm}{2(R+h)}$$

45. Ans (2)

$$\text{Total energy} = -\frac{GMm}{2(R+h)} - \frac{GMm}{2(R+h)},$$

$$g = \frac{GM}{R^2} \Rightarrow g = \frac{GM}{R^2}$$

$$E = -\frac{gR^2m}{2(R+h)} = -\frac{gR^2m}{2(R+h)} \quad h = R$$

$$E = -\frac{mgR^2}{2 \times 2R} = -\frac{mgR}{4}$$

$$E = -\frac{mgR^2}{2 \times 2R} = -\frac{mgR}{4}$$